Astronomy Notes

Contents

1 Fundamentals of Astronomy and Astrophysics **1**
1.1 The Observer's Sky and Coordinate Systems 2
1.2 Magnitudes, Flux, and Luminosity 2
1.3 Spectra and Stellar Temperatures 3
1.4 The Cosmic Distance Ladder 3

2 Stars: Structure and Evolution **3**
2.1 The Hertzsprung-Russell (HR) Diagram 3
2.2 Stellar Structure and Nucleosynthesis 4
2.3 Evolutionary Paths 4

3 Compact Objects: The Undead Stars **4**
3.1 White Dwarfs 4
3.2 Neutron Stars and Pulsars 5
3.3 Black Holes 5
3.4 X-ray Binaries and Accretion 5

4 Galaxies and the Large Scale Structure **6**
4.1 The Milky Way Galaxy 6
4.2 Galactic Classification and AGN 6
4.3 Gamma-Ray Bursts (GRBs) and Mergers 6

1 Fundamentals of Astronomy and Astrophysics

Astrophysics is the branch of astronomy that employs the principles of physics and chemistry to ascertain the nature of astronomical objects, rather than merely their positions or motions in space. To understand the universe, we must first establish the fundamental observational techniques and theoretical frameworks used to measure and classify celestial bodies.

1.1 The Observer's Sky and Coordinate Systems

To map the sky, astronomers project the positions of celestial objects onto an imaginary sphere surrounding the Earth, known as the Celestial Sphere. Two primary coordinate systems are utilized:

- **Altazimuth System (Local):** This system depends on the observer's location and time. It uses *Altitude* (angle above the horizon) and *Azimuth* (angle along the horizon, measured clockwise from North). While intuitive, coordinates change continuously due to Earth's rotation.
- **Equatorial System (Universal):** This system is fixed relative to the stars. It is analogous to Earth's latitude and longitude but projected onto the sky.
  - *Declination (δ):* The angular distance North or South of the Celestial Equator (analogous to latitude), ranging from $-90°$ to $+90°$.
  - *Right Ascension (α):* The angular distance measured eastward along the Celestial Equator from the Vernal Equinox (analogous to longitude), usually measured in hours, minutes, and seconds (0^h to 24^h).

1.2 Magnitudes, Flux, and Luminosity

The brightness of a star is quantified using the magnitude system, a logarithmic scale originating from ancient Greece (Hipparchus). The human eye perceives brightness logarithmically, which is formalized by Pogson's Equation [7]:

$$m_1 - m_2 = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right) \tag{1}$$

where m is the apparent magnitude and F is the measured flux (energy per unit area per unit time).

To compare the intrinsic brightness of stars, we use **Absolute Magnitude** (M), defined as the apparent magnitude a star would have if it were located at a standard distance of 10 parsecs (10 pc). The relationship between apparent magnitude, absolute magnitude, and distance d (in parsecs) is the distance modulus:

$$m - M = 5 \log_{10} d - 5 = 5 \log_{10} \left(\frac{d}{10 \text{ pc}} \right) \tag{2}$$

Luminosity (L) is the total energy emitted by a star per unit time (measured in Watts or solar luminosities $L_\odot$). It is related to flux by the inverse-square law:

$$F = \frac{L}{4\pi d^2} \tag{3}$$

1.3 Spectra and Stellar Temperatures

Light is the primary carrier of astronomical information. By dispersing light into a spectrum, we can determine a star's temperature, composition, and velocity. Stars are closely approximated as blackbodies. According to **Wien's Displacement Law**, the peak wavelength ($\lambda_{\max}$) of a blackbody's emission is inversely proportional to its temperature:

$$\lambda_{\max} T = 2.898 \times 10^{-3} \text{ m K} \tag{4}$$

Furthermore, the total flux emitted per unit surface area of a blackbody is given by the **Stefan-Boltzmann Law**:

$$F_{\text{surface}} = \sigma T^4 \tag{5}$$

Thus, a star's total luminosity depends on its radius R and effective surface temperature T:

$$L = 4\pi R^2 \sigma T^4 \tag{6}$$

Stellar spectra also contain absorption lines (following Kirchhoff's Laws), created when cooler gas in the stellar atmosphere absorbs specific frequencies of light. The spectral lines allow for precise chemical abundance analysis and radial velocity measurements via the Doppler Effect.

1.4 The Cosmic Distance Ladder

Measuring distances is a fundamental challenge in astrophysics. Astronomers use a "ladder" of overlapping methods:

1. **Trigonometric Parallax:** For nearby stars, observing the apparent shift of a star against background stars as Earth orbits the Sun. $d(\text{pc}) = 1/p(\text{arcsec})$.

2. **Standard Candles:** Objects with known intrinsic luminosity.

    - *Cepheid Variables:* Pulsating stars whose period is directly related to their intrinsic luminosity (Leavitt Law [4]). Useful for nearby galaxies.
    - *Type Ia Supernovae:* Thermonuclear explosions of white dwarfs. They have a consistent peak luminosity, allowing distance measurements across billions of light-years.

2 Stars: Structure and Evolution

2.1 The Hertzsprung-Russell (HR) Diagram

The HR Diagram is the most important classification tool in stellar astrophysics. It plots a star's Luminosity (or Absolute Magnitude) against its Surface Temperature (or Spectral

Type: O, B, A, F, G, K, M). Stars do not populate this diagram randomly. Over 90% of stars fall on a diagonal band called the **Main Sequence**, where they are fusing hydrogen into helium in their cores. Other distinct regions include the *Red Giants* and *Supergiants* (cool but highly luminous due to immense radii) and *White Dwarfs* (hot but extremely faint, being very small).

2.2 Stellar Structure and Nucleosynthesis

A star is a self-gravitating sphere of plasma in **Hydrostatic Equilibrium**. The inward pull of gravity is perfectly balanced by the outward pressure of hot gas and radiation [1, 5]. The energy that sustains this pressure is generated via nuclear fusion in the core. For Main Sequence stars like our Sun, hydrogen is fused into helium primarily via the *Proton-Proton Chain* or the *CNO Cycle* (in more massive stars).

When hydrogen in the core is depleted, the core contracts and heats up, while the outer layers expand and cool, turning the star into a Red Giant. If the star is massive enough, core temperatures will rise sufficiently to fuse helium into carbon via the *Triple-Alpha Process*, and subsequently heavier elements up to iron.

2.3 Evolutionary Paths

A star's evolutionary fate is almost entirely determined by its initial mass:

- **Low-to-Intermediate Mass Stars ($M < 8M_\odot$):** These stars undergo Red Giant phases, pulsate, and eventually eject their outer layers to form beautiful *Planetary Nebulae*. The exposed, hot core remains as a White Dwarf.
- **High-Mass Stars ($M > 8M_\odot$):** These stars evolve rapidly, burning through successive nuclear fuels (Carbon, Neon, Oxygen, Silicon) in concentric shells, creating an "onion-like" internal structure. Fusion stops at Iron, as fusing Iron absorbs energy rather than releasing it. The core collapses catastrophically, rebounding to blow the star apart in a brilliant *Core-Collapse Supernova (Type II)*.

3 Compact Objects: The Undead Stars

When a star exhausts its nuclear fuel, it leaves behind a dense remnant. The nature of this remnant depends on the core's mass at the time of death.

3.1 White Dwarfs

White Dwarfs are the remnants of low-mass stars. They are extremely dense (typically packing the mass of the Sun into a volume the size of the Earth). They are no longer generating energy through fusion. Gravity is balanced not by thermal pressure, but by

Electron Degeneracy Pressure. According to the Pauli Exclusion Principle, fermions (like electrons) cannot occupy the same quantum state. When compressed, electrons are forced into higher energy states, generating a purely quantum mechanical pressure. However, Subrahmanyan Chandrasekhar proved that this pressure has a limit. The **Chandrasekhar Limit** [2] is approximately $1.4M_\odot$; a white dwarf exceeding this mass will collapse.

3.2 Neutron Stars and Pulsars

If the collapsing core's mass is between roughly $1.4M_\odot$ and $3M_\odot$, electron degeneracy is overcome. Protons and electrons merge to form neutrons, emitting neutrinos. The core becomes a gigantic atomic nucleus held up by **Neutron Degeneracy Pressure** (and strong nuclear forces). Neutron stars are incredibly dense (a city-sized sphere containing more than a solar mass). Due to conservation of angular momentum and magnetic flux during the collapse, they rotate furiously and possess immense magnetic fields. Beams of radiation emitted from their magnetic poles sweep across space like a lighthouse. When these beams intersect Earth, we detect them as highly regular pulses of radio waves: a **Pulsar**.

3.3 Black Holes

If the remnant core exceeds the Tolman-Oppenheimer-Volkoff limit ($\sim 3M_\odot$) [6], no known physical force can halt the collapse. The mass shrinks to a singularity, a point of infinite density. The gravitational field is so intense that space-time is severely warped. Within a certain boundary, the escape velocity exceeds the speed of light (c). This boundary is the **Event Horizon**, and its radius is the **Schwarzschild Radius** [8]:

$$R_s = \frac{2GM}{c^2} \tag{7}$$

Black holes emit no light themselves but can be detected through their immense gravitational influence on nearby matter, such as in X-ray Binaries.

3.4 X-ray Binaries and Accretion

Many stars exist in binary systems. If one star is a compact object (White Dwarf, Neutron Star, or Black Hole) and its companion expands to fill its Roche Lobe, material will transfer from the companion onto the compact object. Due to angular momentum, the infalling gas forms an **Accretion Disk**. As the gas spirals inward, friction heats it to millions of degrees, causing it to emit brightly in X-rays. If the compact object is a White Dwarf, accumulated hydrogen on its surface can undergo explosive thermonuclear fusion, causing a *Nova*. If the White Dwarf's mass is pushed over the Chandrasekhar limit, the entire star detonates in a *Type Ia Supernova*.

4 Galaxies and the Large Scale Structure

4.1 The Milky Way Galaxy

Our solar system resides in the Milky Way, a barred spiral galaxy containing hundreds of billions of stars [9]. It consists of three main components:

- **The Disk:** A thin, rotating structure containing gas, dust, and young Population I stars (including the Sun). This is the site of active star formation, particularly within its spiral arms.
- **The Bulge:** A dense, roughly spherical central region containing older, redder stars. At the very center lies Sagittarius A*, a Supermassive Black Hole of ~ 4 million $M_\odot$.
- **The Halo:** A vast, spherical region enveloping the galaxy, containing very old, metal-poor Population II stars, Globular Clusters, and an immense, invisible scaffolding of Dark Matter.

4.2 Galactic Classification and AGN

Edwin Hubble classified galaxies by their morphology (the "Tuning Fork" diagram) [3]:

- *Elliptical Galaxies (E0-E7):* Featureless, spherical to elongated systems, largely devoid of gas, containing old stellar populations.
- *Spiral Galaxies (Sa-Sc / SBa-SBc):* Disk galaxies with distinct spiral arms, rich in gas and young stars.
- *Irregular Galaxies:* Lacking distinct structure, often the result of galactic interactions or collisions.

Some galaxies possess an **Active Galactic Nucleus (AGN)**. These centers are incredibly luminous, outshining the entire rest of the host galaxy. The energy source is a Supermassive Black Hole (millions to billions of solar masses) actively accreting massive amounts of gas. Quasars, Blazars, and Seyfert galaxies are all manifestations of AGN viewed under different conditions.

4.3 Gamma-Ray Bursts (GRBs) and Mergers

Gamma-Ray Bursts are the most luminous electromagnetic events in the universe since the Big Bang. They come in two primary populations:

- **Long GRBs (>2 seconds):** Associated with the death of massive, rapidly rotating stars (Collapsars) in distant galaxies. They are closely linked to Type Ic supernovae.

- **Short GRBs (< 2 seconds):** Originating from the catastrophic merger of two compact objects, such as a binary neutron star system or a neutron star and a black hole. These events are also the sources of *Kilonovae* (where heavy elements like gold and uranium are forged) and are primary sources of high-frequency Gravitational Waves.

As galaxies evolve, they frequently collide and merge. The Milky Way, for instance, is currently on a collision course with the Andromeda Galaxy. Over billions of years, these interactions drive the evolution of galaxies from spirals into large ellipticals, shaping the intricate cosmic web of the large-scale structure of the Universe.

References

[1] Bradley W. Carroll and Dale A. Ostlie. *An Introduction to Modern Astrophysics*. 2nd. Cambridge University Press, 2017. ISBN: 978-1108422161.

[2] Subrahmanyan Chandrasekhar. "The Maximum Mass of Ideal White Dwarfs". In: *The Astrophysical Journal* 74 (1931), pp. 81–82. DOI: `10.1086/143324`.

[3] Edwin Hubble. "A Relation between Distance and Radial Velocity among Extra-Galactic Nebulae". In: *Proceedings of the National Academy of Sciences* 15.3 (1929), pp. 168–173. DOI: `10.1073/pnas.15.3.168`.

[4] Henrietta S. Leavitt and Edward C. Pickering. "Periods of 25 Variable Stars in the Small Magellanic Cloud". In: *Harvard College Observatory Circular* 173 (1912), pp. 1–3.

[5] Dan Maoz. *Astrophysics in a Nutshell*. 2nd. Princeton University Press, 2016. ISBN: 978-0691164793.

[6] J. R. Oppenheimer and G. M. Volkoff. "On Massive Neutron Cores". In: *Physical Review* 55.4 (1939), pp. 374–381. DOI: `10.1103/PhysRev.55.374`.

[7] N. R. Pogson. "Magnitudes of Minor Planets". In: *Monthly Notices of the Royal Astronomical Society* 17 (1856), pp. 12–15. DOI: `10.1093/mnras/17.1.12`.

[8] Karl Schwarzschild. "Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie". In: *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften* 1916 (1916), pp. 189–196.

[9] Linda S. Sparke and John S. Gallagher. *Galaxies in the Universe: An Introduction*. 2nd. Cambridge University Press, 2007. ISBN: 978-0521671866.